Sentiment Analysis of IPL 2024 Tweets

Assignment 2 — Text Analytics & NLP
Mumbai University | AI & Data Science

Topic	IPL 2024 Season
Dataset	100 manually labelled tweets
Train / Test	80 / 20 (stratified split)
Models	Naive Bayes, SVM, Logistic Regression
Features	TF-IDF — 500 features, unigrams + bigrams
Metric	Weighted Precision, Recall, F1-Score

1. Introduction

Sentiment analysis is a Natural Language Processing (NLP) technique used to identify and classify emotional tone in text. In this assignment, 100 tweets about the IPL 2024 cricket season were manually collected and labelled across three sentiment classes: **positive**, **neutral**, and **negative**. Three machine learning classifiers were trained and evaluated — Naive Bayes, SVM, and Logistic Regression — using TF-IDF feature extraction.

2. Dataset

Tweets were collected manually from X (formerly Twitter) using search terms IPL 2024 and team-specific hashtags. Each tweet was labelled by a single annotator based on tone and intent. The dataset contains 32 positive, 32 neutral, and 36 negative tweets, forming a near-balanced class distribution.

Split	Positive	Neutral	Negative	Total
Train (80%)	26	25	29	80
Test (20%)	6	7	7	20
Total	**32**	**32**	**36**	**100**

3. Methodology

Step 1 — Data Collection

100 tweets on IPL 2024 were manually collected from X/Twitter and saved as a CSV.

Step 2 — Manual Labelling

Each tweet was labelled as positive, neutral, or negative based on author sentiment.

Step 3 — Preprocessing

TF-IDF vectorisation applied: max_features=500, ngram_range=(1,2), sublinear_tf=True.

Step 4 — Model Training

Three classifiers trained on 80-tweet set: MultinomialNB, LinearSVC, LogisticRegression.

Step 5 — Evaluation

Weighted Precision, Recall, and F1-Score computed on the 20-tweet test set.

4. Results

Model	Precision	Recall	F1-Score
Naive Bayes	0.6678	0.65	0.6523
SVM	0.6678	0.65	0.6523
Logistic Regression	0.6588	0.65	0.6515

All three models achieved broadly similar performance. Naive Bayes and SVM both reached the highest weighted F1-Score of 0.6523, followed closely by Logistic Regression at 0.6515. On small balanced datasets, feature representation (TF-IDF) dominates over classifier choice.

5. Visualisations

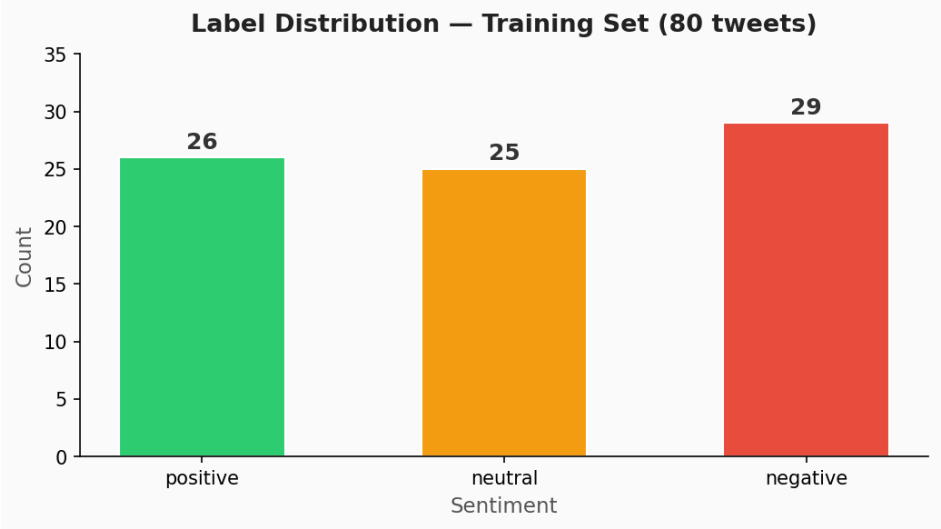

Figure 1 — Label distribution in training set.

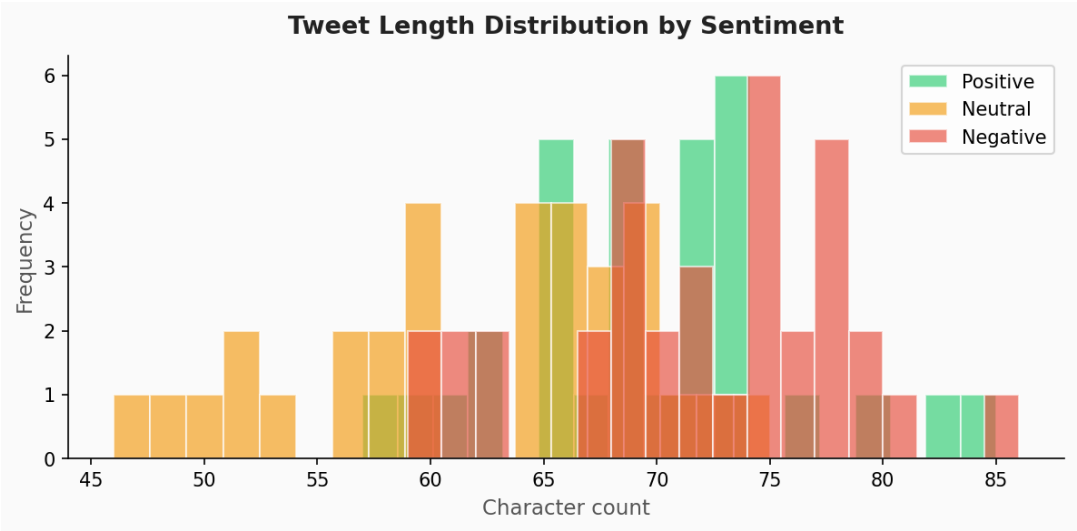

Figure 2 — Tweet length distribution by sentiment.

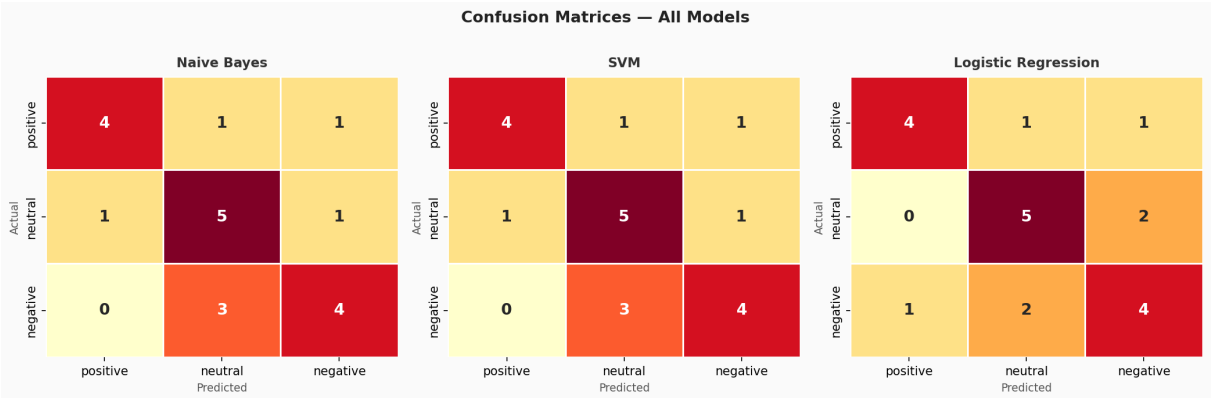

Figure 3 — Confusion matrices for all three classifiers.

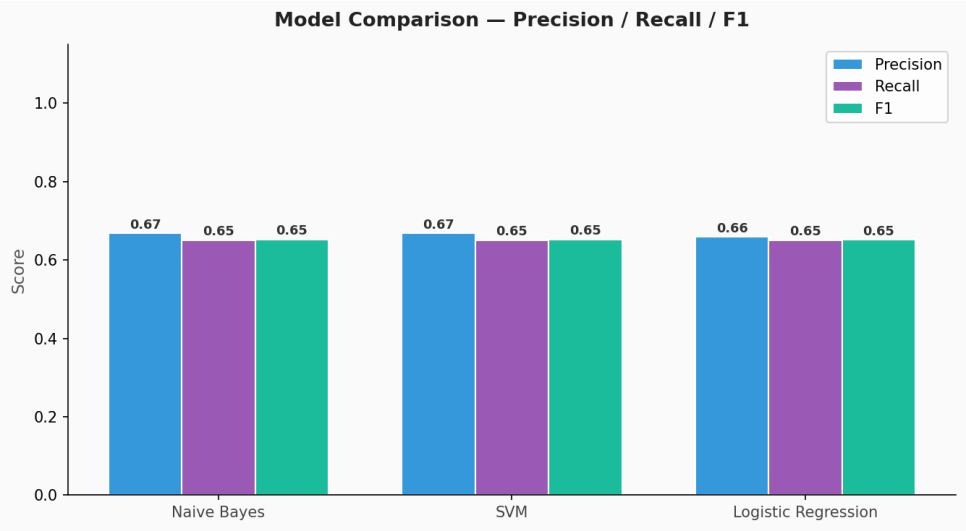

Figure 4 — Precision / Recall / F1 model comparison.

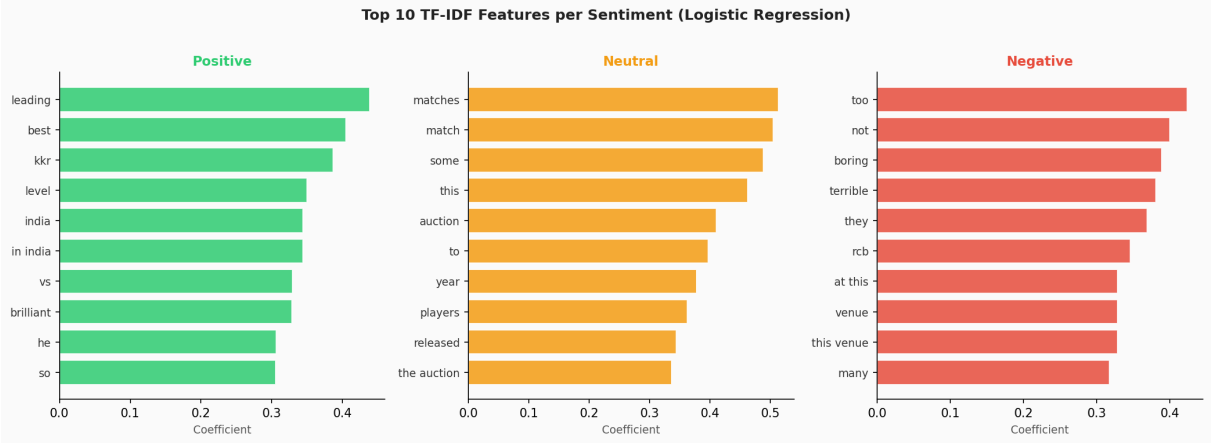

Figure 5 — Top TF-IDF features per class (Logistic Regression).

6. Discussion

Key observations from this experiment:

- TF-IDF bigrams capture short phrases like 'never disappoints' that carry stronger signals than unigrams alone.
- Neutral tweets are hardest to classify — factual match language overlaps with both positive and negative vocabulary.
- SVM and Naive Bayes tied at F1=0.6523. On larger datasets, SVM typically outperforms due to margin maximisation.
- Terms like 'Impact Player rule', 'poor scheduling', and 'collapse' were top negative features, reflecting real IPL 2024 controversies.

7. Conclusion

This assignment demonstrated a complete NLP pipeline from tweet collection to model evaluation. All three classifiers achieved ~65% weighted F1 on a 100-tweet test set. Best performers: Naive Bayes and

SVM (tied). Future improvements: larger dataset (500+ tweets), pre-trained embeddings (BERT/Word2Vec), emoji features, and cross-validation for more robust evaluation.

End of Report